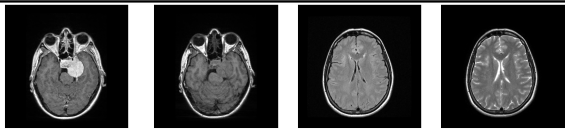
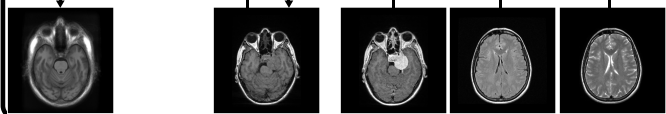


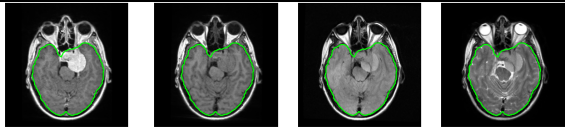
t_0 NIFTI+Bias Field Correction+1mm³ Interpolation

ATLAS Co-registration: t1n-based Composite Transforms

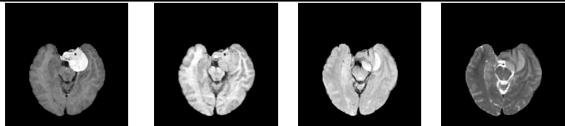
$$T_{m_k \rightarrow \text{atlas}} = T_{\text{ref} \rightarrow \text{atlas}} \circ T_{m_k \rightarrow \text{ref}}^{\text{intra}}, \quad T_{m_k \rightarrow \text{ref}}^{\text{intra}} = T_{\text{affine}}^{(k)} \circ T_{\text{rigid}}^{(k)}, \quad m_k \in \mathcal{M}_t \setminus \{m_{\text{ref}}\}$$



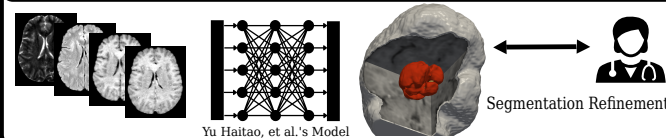
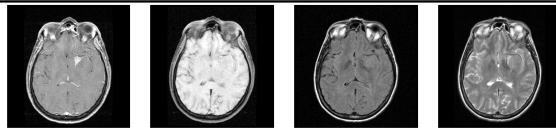
Skull Stripping (Accurate HD-BET on t1c)



Longitudinal Registration (reference)

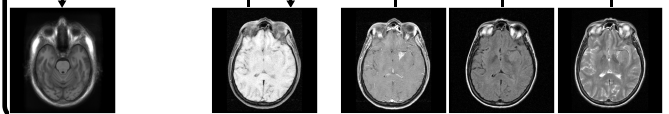


Pre-Segmentation (Yu Haitao, et al. 2025)

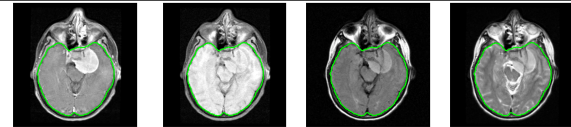
 t_i NIFTI+Bias Field Correction+1mm³ Interpolation

ATLAS Co-registration: t1n-based Composite Transforms

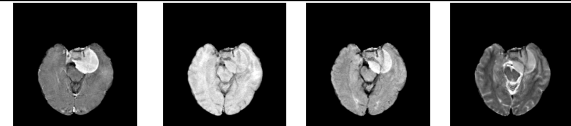
$$T_{m_k \rightarrow \text{atlas}} = T_{\text{ref} \rightarrow \text{atlas}} \circ T_{m_k \rightarrow \text{ref}}^{\text{intra}}, \quad T_{m_k \rightarrow \text{ref}}^{\text{intra}} = T_{\text{affine}}^{(k)} \circ T_{\text{rigid}}^{(k)}, \quad m_k \in \mathcal{M}_t \setminus \{m_{\text{ref}}\}$$



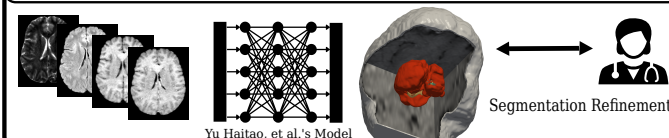
Skull Stripping (Accurate HD-BET on t1c)



Longitudinal Registration (moving)



Pre-Segmentation (Yu Haitao, et al. 2025)



$$T_{t_i \rightarrow t_0}^{\text{long}} : \hat{I}_m^{(t_i)} = I_m^{(t_i)} \circ T_{t_i \rightarrow t_0}^{\text{long}^{-1}}, \quad \forall m \in \mathcal{M}$$